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CURRENT OPINION

Is Fatigue a Risk Factor for Anterior Cruciate Ligament Rupture?

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Abstract

Neuromuscular fatigue is a commonly accepted risk factor for anterior cruciate ligament (ACL) injury. It has been proposed that fatigue leads to transient reductions in muscle strength, and deleterious changes in lower limb kinematics and kinetics, during potentially hazardous tasks such as cutting or landing. The purpose of this clinical commentary is to (1) highlight the complexity of fatigue; (2) discuss the theoretical basis by which it is thought to contribute to ACL injury; and (3) critically discuss the evidence underpinning this hypothesis. Despite a significant amount of research, none of the published fatigue protocols appear to have any consistent effect on any lower limb kinematic or kinetic variables known to increase ACL injury risk. On the contrary, fatigued athletes appear to land with greater peak knee and hip flexion angles, and lower landing forces than unfatigued athletes—all of which are considered favourable movement strategies for reducing ACL loading. These data support recent analyses demonstrating no relationship between player workload in training and competition and the occurrence of ACL injury in sport.

Key Points

The broad scope with which the term 'fatigue' is used causes confusion and there is a lack of interdisciplinary consensus on what it encompasses.

Fatigue is a commonly cited risk factor for anterior cruciate ligament (ACL) injury; however, few prospective studies have quantified this parameter in any meaningful way.

No published fatigue protocols have any consistent effect on any lower limb kinematic or kinetic variables known to increase ACL injury risk.

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1 Introduction

Anterior cruciate ligament (ACL) ruptures are among the most debilitating injuries in landing-, pivoting- and jumping-based sports [1, 2]. It is well-recognised that the aetiology of ACL injury is multifactorial and likely results from the interaction of several modifiable and non-modifiable risk factors [3]. One potentially modifiable factor that has received considerable attention in recent years is the role of neuromuscular fatigue [4, 5]. Proponents of the 'fatigue theory' [6, 7] argue that fatigue leads to acute reductions in muscle strength, altered patterns of lower limb muscle activation, changes in hip and knee kinematics, and increased ground reaction forces during landing or cutting—all of which are hypothesised to increase the likelihood of non-contact ACL injury. However, despite its near universal acceptance, few prospective studies have quantified fatigue in any meaningful way and there are few to no data to indicate that fatigue is indeed a risk factor for ACL injury [4, 5]. Furthermore, none of the neuromuscular training interventions that have proven successful in reduction of ACL injury rates [8] target fatigue-resistance, and there is no evidence to suggest that these ACL injury reduction programmes should be altered to focus on this parameter. This clinical commentary aims to (1) highlight the complexity of fatigue; (2) discuss the theoretical basis by which it is thought to contribute to ACL injury; and (3) critically discuss the evidence underpinning this hypothesis.

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2 What is Fatigue?

By definition, fatigue refers to a transient exercise-induced reduction in the ability to produce force or power, whether or not the task can be sustained [9]. When a task involves sustenance of a maximal contraction, the rate of fatigue is analogous to the decline in performance. However, when the task involves submaximal contractions—as experienced in most sporting activities—the onset of fatigue may not limit the ability to perform the task. This concept of task specificity is an important development in the field that has emerged over the past century. Most contemporary models of fatigue recognise that the condition can be attributed to both central and peripheral mechanisms [9]. Central fatigue refers to an exercise-induced reduction in the ability to fully voluntarily activate muscle, and is thought to arise as a consequence of both spinal and supraspinal factors. This can include suboptimal neural drive from the motor cortex [10], reduced motor neuron firing rates and inhibition of spinal excitability due to muscle afferent input [11]. In contrast, peripheral fatigue refers to changes at or distal to the neuromuscular junction [12] and occurs primarily due to a reduction in the release of calcium ions from the sarcoplasmic reticulum [13]. Fatigue can be quantified via measurement of exercise-induced changes in maximal voluntary force, maximal voluntary shortening velocity or power [9]; however, more specific tests are required to determine the extent to which any deficits are centrally or peripherally mediated.

Despite a significant amount of research over the past 100 years, little is known about the impact of fatigue on human performance and injury risk [14]. One possible explanation for the poor rate of progress is the broad scope with which the term is used, and the lack of interdisciplinary consensus on what it encompasses. For example, fatigue is often used to describe both the gradual decline in muscular force or power experienced during exercise [9], and the point of volitional task failure (i.e. exhaustion) [15]—two phenomena that likely arise from completely different physiological mechanisms. According to other authors, fatigue can refer to deficits in cognitive ability [5], reduced motor function [16] or even perceived changes in homeostatic control systems [17]. In medical practice, fatigue is apparently the most common patient-reported symptom [18] but is usually entirely unrelated to deficits in muscle performance. The problem with such fragmentary interpretations of fatigue is that it renders the parameter virtually unmeasurable; while some researchers assess changes in muscle force, others report alterations in electromyographical (EMG) activity or subjective measures of mental acuity. As a consequence, it is often difficult to determine the extent to which an individual is fatigued during exercise and, by extension, the impact of a fatiguing task on a pre-defined outcome measure.

3 Theoretical Basis for an Effect of Fatigue on Anterior Cruciate Ligament (ACL) Injuries

The role of fatigue in ACL injuries is predicated on the notion that fatigued muscles absorb less energy before failure than non-fatigued muscles [19]. It is hypothesised that this fatigue leads to reductions in muscle strength and consequently deleterious alterations in hip, knee and ankle kinematics and kinetics, which may serve to increase strain on the ACL during potentially hazardous movements such as cutting or landing [6, 7, 20]. Indeed, evidence suggests that excessive dynamic knee valgus [21, 22], poor trunk control (i.e. lateral flexion) [21–23], excessive dorsiflexion [24], and higher ground reaction forces [22] may all elevate the risk for ACL tears during these types of manoeuvres. However, while this theoretical framework may be intuitive, it is equally defensible to suggest that fatigued athletes would actually move more slowly and produce less ground reaction forces than non-fatigued athletes. Such a strategy would be expected to reduce knee joint torques, which may in fact confer a protective effect against ACL injury. To date, there is little consensus on which argument is correct.

4 Do ACL Injuries Occur Later in a Game or Season?

The most commonly used 'measure' of fatigue in sport is the incremental workload related to playing time [5]. While this is not a true index of fatigue, it is generally agreed that athletes who spend more time training or competing experience greater workloads and, by extension, more fatigue, than those who spend less time training or competing. By this logic, if fatigue were a risk factor for ACL rupture, then injuries would occur at a greater frequency later in the game or season. However, contrary to popular opinion, a recent meta-analysis [25] reported no difference in ACL injury occurrence during the first or second halves (odds ratio [OR] = 0.43, 95% confidence interval [CI] 0.07–2.56), or in the first in comparison with the fourth quarter of games (OR = 0.84, 95% CI 0.22–2.68). In this analysis, the authors also demonstrated no difference in ACL rates in the first compared with the second halves of a season (OR = 1.23, 95% CI 0.44–3.46). Whilst there are limitations to such datasets, overall they do lead one to question the role of fatigue in ACL injury.

4.1 Impact of Fatigue Protocols on Biomechanical Risk Factors for ACL Injury

A number of studies have examined the acute effects of a fatigue protocol on lower limb kinematics and kinetics [4].

Most of these studies have employed three-dimensional motion analysis and force plate systems to assess athletes when landing from a hop or jump or performing a cutting manoeuvre. However, despite frequent use of the term 'fatigue' as an outcome measure in these prospective risk factor studies, considerable heterogeneity exists in how different authors have stimulated and measured this parameter. Some studies have exposed athletes to peripheral fatigue protocols that targeted specific lower limb muscles or regions, whereas others employed more general fatigue protocols [4, 5]. Peripheral fatigue protocols range from dynamic lower limb tasks (i.e. squatting, jumping and landing) [26–29] to isolated exercise for the knee extensors and flexors [30–36], hip extensors [37], abductors [38, 39] and rotators [40], or plantar flexors [40, 41] and dorsiflexors [41]. General fatigue protocols have included multi-modal circuits (typically consisting of jumping, squatting and agility tasks) [42, 43], cycling [44], intermittent [45] or prolonged running [46, 47], or simulated 45–60 min football matches [48, 49]. In almost all of the aforementioned studies, fatigue was defined and measured arbitrarily. For example, studies using isokinetic dynamometry define fatigue as a decrement in torque ranging anywhere from 30% [30] to 80% [40]. Similarly, those using jump height as an index of fatigue accepted decrements that ranged from 10% [26] to 20% [42]. Furthermore, a significant number of published studies reported no objective measure of fatigue whatsoever [16, 47, 49–52]. Instead, authors seemed content to assume that athletes had reached a fatigued state based on a set number of repetitions or the duration of the exercise bout. Such an approach neglects to consider that the capacity to tolerate a given workload can vary significantly between athletes based on a number of factors. As a consequence, it is difficult to determine the extent to which athletes in many of these studies were actually fatigued.

4.2 Lower Limb Kinematics

A recent meta-analysis of 20 studies [5] reported that fatigued male and female athletes exhibit significantly greater peak knee flexion (effect size [ES] = −1.27, 95% CI −1.98 to −0.56) and hip flexion angles (ES = −0.48, 95% CI −0.8 to −0.17) during unilateral drop landings. The same analysis reported that athletes displayed a significant decrease in peak knee flexion during the initial contact phase of sidestep cutting (ES = 0.48, 95% CI 0.11–0.84) and a single leg hop for distance (ES = 0.84, 95% CI 0.34–1.34). However, it is important to acknowledge that in this latter analysis, none of the three included studies [52–54] that examined side-stepping reported a significant effect for fatigue on knee flexion during initial contact. Further, the meta-analysis results for single leg hopping for distance were comprised of only two studies [32, 33], which reported conflicting results. A more recent, well-controlled laboratory study [48] reported that athletes exhibit greater peak knee flexion angles during the initial contact phase of side-stepping and a crossover-cutting manoeuvre following a simulated Australian Rules Football match. Studies that assessed ankle kinematics have consistently reported no effect of fatigue on dorsiflexion range of motion or inversion/eversion angles [4]. With respect to frontal plane kinematics, several studies have demonstrated no effect of fatigue on peak knee valgus and hip abduction angles during single leg drop vertical jumps and single leg drop landings [6, 7, 26, 55]. In fact, some data [6] indicate that fatigued men and women display significantly decreased peak knee valgus values during the initial contact phase of a single leg stop jump.

4.3 Lower Limb Kinetics

Increased ground reaction forces are another commonly cited risk factor for ACL injury [22] and a significant number of studies have examined the effect of fatigue on this parameter. According to a recent systematic review [4], 90% of these published investigations have failed to observe an increase in landing forces following a fatigue protocol. Further, the two studies [43, 51] that did identify an increase in ground reaction forces both observed small or trivial ESs. When considering knee joint loading, a recent systematic review and meta-analysis [5] demonstrated no significant effect of fatigue on peak knee abduction moments (ES = 0.28, 95% CI −0.18 to 0.74). Further, Kernozek and colleagues [7] reported that men and women displayed significantly lower knee extension moments following a fatigue protocol.

4.4 Lower Limb Muscle Activation

Several investigations have attempted to map the surface EMG (sEMG) activity of lower limb muscles during various athletic tasks before and after fatigue [16, 33, 51]. While sEMG is not a measure of fatigue, it is hypothesised that central fatigue leads to a reduction in voluntary drive [10], which would be reflected by a decrease in the sEMG amplitude overlying the target muscle. In particular, reduced hamstring relative to quadriceps activation during dynamic knee loading may be associated with lower knee flexion moments, greater knee valgus moments and increased ground reaction forces [3], and one study has observed with sEMG that this pattern of muscle activation is associated with an increased risk of future ACL injury [56]. However, it is important to recognise that the sEMG amplitude is heavily influenced by the degree to which motor unit firing is synchronous [57], which also increases during fatigue, so the results of these types of investigations should be interpreted with caution. Another major limitation of sEMG is its susceptibility to cross-talk from neighbouring muscles, which makes it

difficult to reliably discriminate between closely approximated muscles, especially in the lower limbs [58]. Unsurprisingly, sEMG studies of lower limb muscles before and after fatigue protocols have yielded inconsistent results [4]. For example, Zebis and colleagues [16] reported a reduction in hamstring sEMG amplitude following a simulated soccer match; however, Pappas and colleagues [51] reported no change in this parameter following a fatiguing jumping protocol. Similarly, Ortiz and colleagues [44] reported no change in quadriceps sEMG activity after a Wingate cycle test, whereas Orishimo and colleagues [33] reported an increase in quadriceps sEMG after a fatiguing step-up protocol. The results from studies that assessed plantar flexor activity are also inconsistent [4]. Given the complex nature of fatigue [10] and the methodological concerns associated with sEMG [57, 58], researchers should avoid making inferences about the fatigability of muscles based on this measure alone. Future work, employing direct measures of voluntary activation (i.e. twitch interpolation), is needed to better understand the impact of fatiguing exercise on the magnitude of lower limb muscle activity. Assessment techniques with high spatial resolution, such as functional magnetic resonance imaging [59], may also be used to characterise the effects of fatigue on the 'patterns' of lower limb activation during high-risk activities.

5 Impact of Fatigue on Cognition and Decision-Making

A recently proposed taxonomy [14] suggests that fatigue should be considered as a disabling symptom, derived from the interaction of two interdependent attributes: performance fatiguability and perceived fatiguability. Such a framework considers that fatigue is not modulated by alterations in central and peripheral mechanisms alone, but also by a host of subjective psychological factors. In complex and cognitively demanding situations (i.e. high-risk sports), these factors may interact with and influence the ability of an athlete to make rapid decisions and to react to external stimuli; theoretically, this may lead to deleterious movement strategies that contribute to increased strain on the ACL [5]. While this type of conceptual framework may be intuitively appealing, it has not yet been the subject of rigorous scientific inquiry. One possible method for testing this theory is to examine the movement strategies of fatigued athletes in response to planned and unplanned (i.e. where the athlete must react to an unanticipated stimulus) tasks. However, despite a significant amount of research exploring these types of responses, no consistent differences have been observed between reactive and planned tasks for almost any lower limb kinematic or kinetic variables [4]. The one exception is an isolated finding for knee flexion, which appears to be consistently reduced in reactive but not planned tasks [4]. Future studies are needed to determine if fatigue-induced alterations in cognition and decision-making contribute to deleterious movement strategies in more complex sporting situations; however, based on current evidence, this seems unlikely.

6 Do current Neuromuscular Training Interventions Improve Fatigue-Resistance?

A recent meta-analysis of meta-analyses [8] provided robust evidence that ACL injury prevention programmes reduce the risk of all ACL injuries by half and non-contact ACL injuries by two-thirds in female athletes. Given the multifactorial nature of these neuromuscular training programmes, it is difficult to determine the active ingredient(s) in the interventions. For example, some include training directed at improvement of balance [60], stability or technique [61], whereas others focus on strengthening and plyometric exercises [62] and many target a combination of these variables [63, 64]. However, by conventional standards, none of the training programmes employed in these studies appear to target improvements in fatigue-resistance of lower limb muscles. Most interventions are conducted over a short duration (typically < 20 min), often in a warm-up period, and none involve prescriptions consistent with improving general or peripheral muscular endurance. Furthermore, as far as the authors are aware, none of the published interventions have attempted to measure the effect of training on the fatigability of athletes in any task.

7 Conclusion

It is generally agreed that reduced knee and hip flexion, increased knee abduction angles and moments, and increased ground reaction forces during landing are all risk factors for ACL injuries [3]. However, as highlighted in this clinical commentary, and in recent systematic reviews [4, 5], none of the published fatigue protocols appear to have any consistent effect on these kinetic and kinematic variables. On the contrary, fatigued athletes appear to land with greater peak knee and hip flexion angles [5] and lower landing forces [4] than unfatigued athletes—both of which are considered favourable movement strategies for reducing ACL loading. While not a true measure of fatigue, a recent meta-analyses also demonstrate no effect of playing time on ACL injury rates [25], so there is little reason to suspect that these injuries are influenced by player workload. Finally, currently established neuromuscular training programmes have proven effective in reducing ACL injury rates by at least 50%, and there is no to minimal evidence to support the

recommendation [29] that these primary or secondary ACL risk reduction interventions should be altered to include more specific fatigue-resistance training. Future research examining the effect of fatigue on ACL injuries may benefit from establishing a clear, unambiguous definition of fatigue and a consistent means of measuring it.

Compliance with Ethical Standards

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